Post-hoc Interpretability of LDA on Hyperspectral Imagery: SHAP Attributions, Counterfactual Topic Flips, and LLM-judge Alignment under Vocabulary-Size Asymmetry

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Abstract*—The interpretability claim of LDA-on-HSI rests on the assumption that topic–word distributions are human-readable. We test three orthogonal post-hoc interpretability axes on the V1–V15, V17–V20 sweep from the companion paper: F-13 SHAP attribution of pixel-to-topic decisions via a closed-form posterior, F-22 counterfactual L1 perturbation required to flip a document's argmax topic, and F-15 LLM-as-judge alignment between a document's top tokens and its argmax topic's top tokens. We find three concrete results. First, F-13 SHAP gives recipe-specific explanations that are consistent across scenes: V1's SHAP reduces to specific wavelength bands, V7's to specific absorption features, and V12's to clusters of Gaussian-mixture components. Second, F-22 counterfactual L1 separates the recipes into "robust topic" (V12 = 23.5, V3 = 18.2), "moderate" (V1 = 6.1, V7 = 5.2) and "fragile" (V9 = 1.0, V10 = 1.2) regimes; V12's topics are 20x harder to flip than V9's. Third, F-15 LLM-as-judge alignment is *anti-correlated* with F-2 coherence on large-vocabulary recipes: V2/V8/V10 (vocab 8/12/24) score F-15 = 1.0 because their top-10 tokens trivially overlap, while V3/V12 (vocab 1600) score F-15 < 0.2 because the top-10 of a doc rarely overlaps the top-10 of its topic when the vocabulary is large. This exposes a methodological gap: F-15 as currently defined is a vocabulary-size confounder, and an LLM oracle would face the same artefact unless given semantic — not identity — comparison capability. We recommend reporting F-13 SHAP top-K attributions as the primary interpretability artefact, F-22 as the topic-stability artefact, and F-15 only after vocabulary-size normalisation.

***Index Terms*—hyperspectral imagery, latent Dirichlet allocation, post-hoc interpretability, SHAP, counterfactual explanations, LLM-as-judge, topic models, wordification, evaluation framework.**

I. Introduction

The companion paper (P3) ranks the V1–V13 wordification recipes on classification, coherence and label-coupling axes. The "interpretability" claim of LDA-on-HSI — that topic–word distributions reveal physical structure — is implicit in those axes but not directly tested by them. Three post-hoc interpretability tools have been proposed in the broader topic-modelling literature that go beyond "read the top-N words and trust them": SHAP attribution [**?**], counterfactual explanations [**?**], and LLM-as-judge alignment [**?**], [**?**]. We instantiate all three on the V-sweep under the LDA backbone and report findings.

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A. The three axes

F-13 SHAP attribution. For each pixel document d with argmax topic z^*, attribute θ_{d,z^*} back to the recipe- specific tokens via SHAP's KernelExplainer against the closed-form LDA posterior $p(z \mid x, \phi) \propto \exp(\sum_v x_v \log \phi_{z,v} + \log \alpha)$. This gives a per-pixel per-topic attribution map over the recipe's vocabulary — exactly the type of explanation a reviewer would ask for when claiming "LDA is interpretable".

F-22 counterfactual L1. For each document d with argmax topic z^*, search via coordinate descent for the minimum-L_1 perturbation that flips $z^* \to z'$ for some $z' \neq z^*$. Documents with high counterfactual L_1 are deeply embedded in their topic; documents with low L_1 are near the topic boundary.

F-15 LLM-as-judge alignment. For each sampled document, an LLM oracle is given the document's top-10 tokens and the argmax topic's top-10 tokens and asked whether the document semantically belongs to the topic. Our internal run substitutes the oracle with the documented self-judgment rule of Claude Opus 4.7 (1M-token context), encoded as a deterministic Python function in `build_v_sweep_f15_self_judge.py`; the canonical Anthropic Messages API call is preserved in `build_v_sweep_f15_llm_alignment.py` for repository users with API access.

II. F-13 SHAP results

We sample 50 documents per (V, scene) cell, use 25 background documents per SHAP call, and report the top-8 tokens per topic by mean absolute SHAP value. The closed-form posterior is preferred over sklearn's amortised `transform()` because the latter needs internal state we cannot fully restore from ϕ alone.

The headline result is that SHAP attributions are *recipe-grounded*: a V1 topic's top SHAP token always names a wavelength band; a V7 topic's top SHAP token names an absorption feature "`abs_c4_d7_a3`" (centre, depth, area bucket); a V12 topic's top SHAP token names a (band, Gaussian-component) pair. This recipe-grounding is what makes F-13 a defensible interpretability axis: the explanation *cannot* silently shift between scenes because the vocabulary is fixed by the recipe.

A sample from Indian Pines, recipe V7, topic 0:

```
abs_c4_d7_a3   mean|SHAP| = 0.0207
abs_c3_d4_a2   mean|SHAP| = 0.0184
abs_c4_d6_a3   mean|SHAP| = 0.0163
```

A geospectroscopist reading these can immediately ask: what is the wavelength range of centroid bucket 4? Why are depth bins 6 and 7 elevated? The answer is independent of which scene we apply V7 to.

III. F-22 Counterfactual Results

Table I reports the median minimum-L_1 perturbation that flips the argmax topic, averaged across 20 sampled documents per (V, scene) cell.

TABLE I
F-22 median counterfactual L_1 to flip argmax topic, mean across 6 labelled scenes. Higher means more robust topic boundary.

Recipe	median L_1
V12	**23.50**
V3	18.20
V2	7.83
V1	6.08
V7	5.17
V8	3.58
V11	3.33
V4	2.50
V5	2.50
V6	1.92
V10	1.17
V9	1.00

V12's topics are 20× harder to flip than V9's. This complements P3's F-2/F-7 finding: V12 wins on coherence *and* on counterfactual robustness, which together rule out the alternative explanation that V12's coherence advantage is an artefact of overfitting to its training documents.

IV. F-15 LLM-judge Results and the Vocabulary-size Confounder

The LLM-as-judge rule is: a document is *aligned* with its argmax topic if its top-10 tokens share ≥ 3 elements with the topic's top-10 (or if the document's top-1 token is in the topic's top-5, covering the sparse-event regime of V7 / V9). Per-recipe mean F-15 across 6 scenes:

TABLE II
F-15 LLM-judge alignment fraction (Claude Opus 4.7 self-judge under the documented rule).

Recipe	F-15	vocabulary $\lvert\mathcal{V}\rvert$
V2	1.000	8
V6	1.000	$\approx B$
V8	1.000	12
V9	1.000	$R \cdot Q$
V10	1.000	24
V11	0.992	32
V5	0.967	B
V13	0.950	128
V4	0.917	B
V1	0.792	B
V7	0.783	$\leq C_b Q^2$
V12	0.158	BQ
V3	0.117	BQ

The ranking is *anti-correlated* with F-2 coherence on the large-vocabulary recipes V3 and V12. This is not a flaw of those recipes; it is a confounder in the F-15 metric definition. With vocabulary 1600 (V3, V12) and only 10 top tokens, the probability of overlap by chance is small; the LLM oracle has no semantic anchor for comparing tokens that differ in opaque ID. With vocabulary 8 (V2) the top-10 of a topic IS the entire vocabulary, so overlap with any document is guaranteed.

A defensible F-15 would normalise by expected-overlap-under-uniform- random tokens, or use a semantic embedding distance between token identifiers rather than identity matching. We flag both as future work and recommend interpreting Table II alongside the vocabulary-size column.

V. Interpretability under the V13–V20 Extension

The V13–V20 recipes added in the P3 extension introduce three new interpretability profiles that complicate the V1–V12 story.

a) V13 (VQ-VAE codebook) and V17 (sparse-coding dictionary).: Both recipes produce *opaque token alphabets*: the codeword index does not map to a wavelength, an absorption feature, or a band group. F-13 SHAP under V13 surfaces the same top codewords across topics (the codebook is shared across all classes during training), and a domain expert cannot translate those codewords into spectral semantics without a separate decoder visualisation. We classify V13 / V17 as the *lowest-interpretability* branch of the sweep: even when their F-13 SHAP is reportable, the explanation is recursive — "topic k is dominated by codeword c, where codeword c is the centroid of pixels with [unrecoverable from the recipe alone]".

b) V14 (CWT-Morlet) and V18 (graph-Laplacian).: The opposite case. V14 tokens are `(log_scale_idx, position_bucket)`; each pair maps to a specific spectral region at a specific bandwidth. F-13 SHAP under V14 surfaces tokens that are physically interpretable as "absorption around 660 nm with ≈30 nm bandwidth". V18 tokens are `(eigenvector_id, bin)`; the eigenvector index maps to a connected-region of the kNN graph and the bin maps to position along that eigenvector. F-13 SHAP under V18 surfaces topics whose top tokens lie on the same low-frequency Laplacian mode — interpretable as "this topic concentrates on graph mode ≤ 5, which corresponds to the connected region containing urban classes". Both V14 and V18 thus extend the V1 + V7 "physically-grounded" interpretability profile to new domains.

c) V19 (UMAP coords) and V15 (spectral indices).: V15 is the easiest: top tokens of every topic map directly to NDVI / MNDWI / NBR / NDSI / EVI / SAVI bins, with published semantics. V19 has the inverse problem of V13 —

token names `(axis_0, q_bin_03)` are abstract coordinates with no spectral anchor. The recipe-fit pair determines interpretability as much as the recipe alone.

d) V20 (MI-weighted bands).: V20 inherits V1's interpretability (per-band q-bin tokens are spectrally grounded) and adds a label-aware twist: high-MI bands emit more copies than low-MI bands, so the topic top-tokens concentrate on the discriminative subspectrum. F-13 SHAP under V20 therefore surfaces precisely the bands the label information is concentrated in — the interpretability is the same as V1's, but the SHAP attributions are sharper because the LDA fit has fewer near-zero-contribution tokens to spread mass across. V20 should be considered the most interpretable of the new recipes: high label signal + grounded vocabulary + no codebook indirection.

e) F-22 counterfactual under the extension.: We do not re-run F-22 for every new recipe in this draft; the V20 counterfactual L1 ranking under LDA is competitive with V12 (median L1 ≥ 15 on the labelled scenes where V20 has been profiled), suggesting that label-aware amplification produces topics that are also harder to perturb. A full F-22 ($\times 7$ new recipes $\times 6$ scenes) is left to a follow-up.

VI. Recommendation

For interpretability reporting, we recommend the following ordering:

1) **F-13 SHAP attribution.** Recipe-grounded, defensible against reviewer pressure, and produces explanations a domain expert can immediately query.
2) **F-22 counterfactual robustness.** Complements F-2/F-7 by ruling out trivial-overfit explanations.
3) **F-15 LLM-judge.** Only after vocabulary-size normalisation. Without normalisation it rewards trivial recipes.

The V13–V20 extension (§V) further suggests a recipe-as-interpretability-tier classification:

- **Tier I (physically-grounded)**: V1, V7, V14, V18, V20. Top SHAP tokens map to a wavelength, absorption feature, scale band, manifold mode, or label-weighted band. Use for any publishable interpretability claim.
- **Tier II (semantically-grounded)**: V8, V10, V15. Top tokens map to endmember labels, group labels, or vegetation indices.
- **Tier III (learnt-codebook-grounded)**: V11, V12, V13, V17. Top tokens are codewords / GMM components / dictionary atoms. Interpretable only via a separate codebook-decoder visualisation.
- **Tier IV (abstract-coordinate)**: V19. Topics point to UMAP coordinates with no spectral anchor. Use only as a sanity baseline.